ANSWER KEY & MARKING SCHEME SET 1

Chapter 11: **Earth, Moon and the Sun** | Class 7 Science | Maximum Marks: 40

SECTION — A (Objective Type) — Answers — 10 × 1 = 10

Q.	Correct Option	Brief Reasoning
1	**(c)** more extreme	Larger variation.
2	**(b)** more frequent leap years	Larger quarter.
3	**(b)** the Earth always has a shadow but the Moon enters it only during eclipses	Shadow is always there.
4	**(d)** 12 hours	Half.
5	**(b)** the Earth moves around the Sun meanwhile	Extra two days to realign.
6	**(d)** compare dates of nearest distance with seasons	January is nearest.
7	**(b)** 182 days	Half.
8	**(d)** much nearer to the Earth	It appears equally large in the sky.
9	**(d)** incorrect	Phases are due to viewing angles.
10	**(c)** overhead all year	No tilt.

SECTION — B (Very Short Answer) — Answers — 4 × 2 = 8

Q.	Expected Answer / Value Points	Marks
11	Moon reflects only a small fraction (1); Light is weak (1).	1+1
12	Earth rotates from west to east (1); Relative motion makes them seem to move opposite (1).	1+1
13	Sun stays above the horizon longer (1); Tilt effect (1).	1+1
14	Slight offset from the axis (1); It lies close but not exactly (1).	1+1

SECTION — C (Short Answer) — Answers — 4 × 3 = 12

Q.	Expected Answer / Value Points	Marks
15	15° in 1 hour (1); 1° in 4 minutes (1); 15° takes 1 hour (1).	1+1+1
16	Earth's shadow is bigger (1); Moon's shadow small (1); Speeds (1).	1+1+1
17	Study the Sun's corona (1); Test ideas about light (1); Predict future events (1).	1+1+1
	OR: First quarter about 8 March (1); Full moon about 16 March (1); Last quarter about 23 March (1).	1+1+1
18	Light travels through more air (1); Blue scatters (1); Red remains (1).	1+1+1
	OR: About 12 hours each (1); Little change (1); Sun nearly overhead (1).	1+1+1

SECTION — D (Long Answer) — Answers — 2 × 5 = 10

Q.	Expected Answer / Value Points	Marks
19	Fix a stick on the globe (1); Place lamp at the Sun's position (1); Tilt the globe (1); Measure shadows at two latitudes (1); Shadow longer at higher latitudes (1).	1+1+1+1+1
	OR: Regular motions (1); Records of old eclipses (1); Calculations (1); Computer models (1); Accurate predictions (1).	1+1+1+1+1
20	Narrow shadow (1); Path of totality (1); Safe viewing: pinhole or filters (1); Record time and observations (1); Share with class (1).	1+1+1+1+1
	OR: North of east in summer (1); South of east in winter (1); Due east on equinox (1); Tilt (1); Observation by markers (1).	1+1+1+1+1

Evaluator's note: Award step marks as shown; accept any correct alternative method. In internal-choice questions, evaluate only the option attempted. Deduct up to ½ mark for missing units or figures where required.

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ANSWER KEY & MARKING SCHEME SET 2

Chapter 11: **Earth, Moon and the Sun** | Class 7 Science | Maximum Marks: 40

SECTION — A (Objective Type) — Answers — 10 × 1 = 10

Q.	Correct Option	Brief Reasoning
1	**(a)** about 2.5 times	The Earth's shadow is large.
2	**(a)** incorrect	Far side gets sunlight.
3	**(c)** every month	Alignment each month.
4	**(a)** the Sun is overhead	Equinox.
5	**(c)** incorrect	It rotates once per orbit.
6	**(b)** all places where the Moon is above the horizon	Night side.
7	**(c)** sunrise and sunset	Low Sun.
8	**(d)** incorrect	It is in January.
9	**(d)** more common	Appears larger.
10	**(c)** the Sun is higher in the sky in June	Higher Sun.

SECTION — B (Very Short Answer) — Answers — 4 × 2 = 8

Q.	Expected Answer / Value Points	Marks
11	Sun's position changes (1); Earth rotates (1).	1+1
12	Earth's rotation (1); Time difference (1).	1+1
13	Revolution changes the Sun's direction (1); Tilt changes the sunlight pattern (1).	1+1
14	Moon's umbra is narrow (1); Path of totality (1).	1+1

SECTION — C (Short Answer) — Answers — 4 × 3 = 12

Q.	Expected Answer / Value Points	Marks
15	Summer shadow shorter (1); Sun higher (1); Winter Sun lower (1).	1+1+1
16	Stick shadow at noon (1); Pole star near north (1); Compass (1).	1+1+1
17	Sun rises and sets daily (1); Tilt effect small (1); Day about 12 h (1).	1+1+1
	OR: Year is slightly less than 365.25 (1); Century years not divisible by 400 skipped (1); 100 - 3 = 97 (1).	1+1+1
18	Moon blocks the light (1); Umbra (1); Temperature drops (1).	1+1+1
	OR: Altitude (1); Thin air holds less heat (1); Temperature falls with height (1).	1+1+1

SECTION — D (Long Answer) — Answers — 2 × 5 = 10

Q.	Expected Answer / Value Points	Marks
19	Earth rotates: day (1); Moon revolves: month (1); Earth revolves: year (1); Position: phases (1); Alignment: eclipses (1).	1+1+1+1+1
	OR: Earth's shadow causes eclipses only (1); Phases by viewing angle (1); Ball and lamp show lit part (1); Move around (1); Conclude (1).	1+1+1+1+1
20	6 normal years: 365 each (1); 2 leap years: 366 each (1); Total 2922 days (1); Months vary (1); February 29 in leap years (1).	1+1+1+1+1
	OR: Lamp as Sun (1); Globe as Earth (1); Ball as Moon (1); Ball between lamp and globe: solar eclipse (1); Globe between: lunar eclipse (1).	1+1+1+1+1

Evaluator's note: Award step marks as shown; accept any correct alternative method. In internal-choice questions, evaluate only the option attempted. Deduct up to ½ mark for missing units or figures where required.

ANSWER KEY & MARKING SCHEME SET 3

Chapter 11: **Earth, Moon and the Sun** | Class 7 Science | Maximum Marks: 40

SECTION — A (Objective Type) — Answers — 10 × 1 = 10

Q.	Correct Option	Brief Reasoning
1	**(c)** seasons well	Based on revolution.
2	**(c)** all of its surface over a month	Different faces.
3	**(d)** shortest on equinoxes	Sun overhead.
4	**(b)** time taken for one rotation	Rotation unaffected.
5	**(c)** 25	One in four years.
6	**(a)** full moon	Earth is between.
7	**(b)** north (in the northern hemisphere)	Sun is south.
8	**(a)** a day at summer solstice	66.5° N.
9	**(c)** the Sun is highest in the sky	Highest Sun gives shortest shadow.
10	**(a)** new moon, first quarter, full moon, last quarter	Cycle.

SECTION — B (Very Short Answer) — Answers — 4 × 2 = 8

Q.	Expected Answer / Value Points	Marks
11	Dates follow Moon phases (1); Purnima and Amavasya are fixed (1).	1+1
12	Mark a ball and move around (1); Rotation equals revolution time (1).	1+1
13	They depend on the Moon's position, not on the observer (1); All see the same lit part (1).	1+1
14	Sun reaches meridians at different times (1); Rotation of Earth (1).	1+1

SECTION — C (Short Answer) — Answers — 4 × 3 = 12

Q.	Expected Answer / Value Points	Marks
15	Winter (1); Short days (1); Slanting rays (1).	1+1+1
16	Moon is large and nearby (1); Venus is small and far (1); Sky is bright (1).	1+1+1
17	Months by lunar phases (1); Purnima is the full moon (1); Amavasya is new moon (1).	1+1+1
	OR: Day 12 + 29.5 (1); About 10 February (after 29-30 days) (1); Rounding (1).	1+1+1
18	Heat accumulates (1); Monsoon arrives (1); Hottest around May (1).	1+1+1
	OR: Gravity pulls water (1); Close to Earth (1); Tidal bulge (1).	1+1+1

SECTION — D (Long Answer) — Answers — 2 × 5 = 10

Q.	Expected Answer / Value Points	Marks
19	Kerala near equator: little change (1); Delhi: distinct seasons (1); Leh: cold due to altitude (1); Tilt influences all (1); Monsoon adds (1).	1+1+1+1+1
	OR: New moon: solar eclipse possible (1); Two weeks later full moon (1); Lunar eclipse (1); Note 14-15 day gap (1); Explain alignment (1).	1+1+1+1+1
20	No Moon light at night (1); No lunar eclipse (1); No solar eclipse (1); No phases (1); No lunar calendar (1).	1+1+1+1+1
	OR: Festivals like Makar Sankranti and Purnima (1); Myths about eclipses (1); Scientific explanation (1); Safe viewing (1); Community education (1).	1+1+1+1+1

Evaluator's note: Award step marks as shown; accept any correct alternative method. In internal-choice questions, evaluate only the option attempted. Deduct up to ½ mark for missing units or figures where required.